Task A2 — System Architecture Diagram

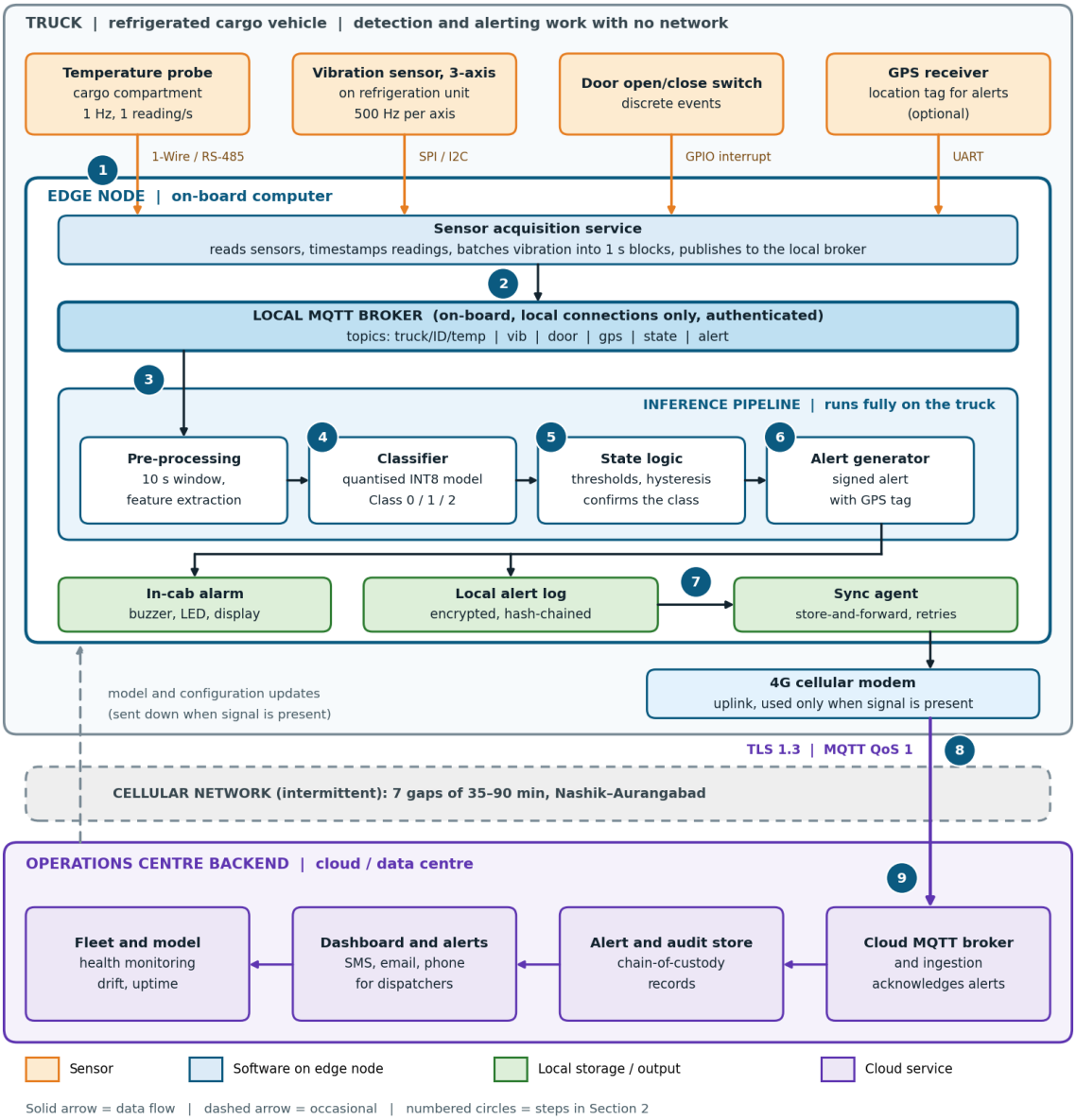
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At a glance

Key fact	Design
Where detection runs	On the truck's edge node; no cloud dependency
Steps that need a network	Only steps 7–9 (uplink and operations centre)
Detection-to-alarm time	≈ 10.15 s (workings in Task A1)
Uplink volume	≈ 0.034 MB per truck per day (Task A1)

1 Architecture diagram

Given: *show all seven required components, from sensors to operations centre.*



Arrows show the logical data flow. Internally, the alert generator publishes each alert on the alert topic, and the log writer and alarm driver subscribe to it.

2 Data flow, step by step

Step	From → To	What moves	Needs network?
1	Sensors → acquisition service	Temperature 1 Hz; vibration 500 Hz × 3 axes; door events; GPS position	No
2	Acquisition service → local broker	MQTT messages: 8 B temperature readings, 1 s vibration blocks of 3,000 B, door events	No
3	Local broker → pre-processing	Subscribes to sensor topics; builds non-overlapping 10 s windows and features	No
4	Pre-processing → classifier	Features in; quantised INT8 model returns Class 0, 1 or 2 with a confidence	No
5	Classifier → state logic	Thresholds (Warning 1–3 °C, Critical >3 °C) and hysteresis confirm the class	No
6	State logic → alert generator → alarm and log	Signed alert (≈ 256 B) written to the local alert log; in-cab alarm triggered	No
7	Local alert log → sync agent	Sync agent reads entries not yet acknowledged	No
8	Sync agent → modem → cellular → cloud	TLS 1.3, MQTT QoS 1; retries until acknowledged	Yes
9	Cloud broker → alert store → dashboard	Ingest, store as chain-of-custody record, notify dispatchers	Yes (cloud side)

Finding: steps 1–6 stay on the truck, so detection and alarms continue through every signal gap. Only steps 7–9 wait for coverage. Even a 3-window confirmation rule (windows complete at 10, 20 and 30 s) would alert in about 30 s, inside the 90 s limit.

3 Behaviour with and without signal

Aspect	Signal present	Signal lost (35–90 min)
Detection and classification	On the truck	Unchanged, on the truck
Driver alert	In-cab alarm	In-cab alarm
Operations centre alert	Sent as soon as the alert is logged	Queued in the local alert log
Records	Logged locally and uploaded	Logged locally; nothing is lost
When signal returns	Not applicable	Queue uploads in order; unacknowledged entries are retried

Finding: the alarm and log form the primary path, and the uplink is a reporting path. A cellular failure therefore cannot stop detection or leave a gap in the record.

4 Component responsibilities

Component	Responsibility	Design choice
Edge node	Runs acquisition, broker, pipeline, log and sync	One on-board computer; hardware justified in a later task
Sensor connections	Feed raw readings to the acquisition service	One dedicated link per sensor (examples: 1-Wire or RS-485, SPI or I2C, GPIO, UART); finalised at hardware selection
Local MQTT broker	Routes every on-truck message	Local connections only, authenticated; sensors and pipeline can change independently
Inference pipeline	Turns readings into Class 0, 1 or 2	Fully on device; quantised INT8 model
Local alert log	Permanent record of alerts and temperature	Append-only, AES-256 encrypted, hash-chained; also holds the 1 Hz temperature log (0.69 MB/day)
Cellular uplink	Moves alerts to the operations centre	Sync agent plus 4G modem; TLS 1.3 with mutual authentication; MQTT QoS 1
Operations centre backend	Receives, stores, notifies, monitors	Cloud MQTT broker, alert and audit store, dispatcher dashboard, fleet and model health monitoring

5 Local MQTT topic design

Topic	Rate	Payload	QoS	Subscribers
truck/ID/temp	1 per second	8 B reading	0	Pipeline, log
truck/ID/vib	1 per second	3,000 B block (500 Hz × 3 axes × 2 B)	0	Pipeline
truck/ID/door	On each event	16 B event	1	Pipeline, log
truck/ID/gps	Periodic	Position	0	Alert generator (location tag)
truck/ID/state	Every 10 s window	Class and confidence	0, retained	In-cab display, status summary
truck/ID/alert	On each alert	≈ 256 B signed message	1	Log writer, alarm driver

Finding: local MQTT traffic never leaves the truck. Only status packets and alerts (about 34 KB per day) cross the cellular link, matching the Task A1 bandwidth figures.

6 Items to confirm

- Compute hardware for the edge node (selected and justified in a later task).
- Sensor models and interfaces; those shown are examples.
- MQTT broker software, and whether the operations centre backend is cloud-hosted or on-premises.
- Hysteresis rule (how many windows confirm a class change), tuned during validation.
- Log retention period, and whether over-the-air model updates are wanted (shown dashed).
- Whether a buzzer, LED or display is already fitted in the cabs.